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1. RECOLLECTIONS ON SUBGROUPS

Definition 1.1 (Subgroup). Let $(G, \cdot, e)$ be a group. A subset $H \subseteq G$ is a *subgroup* if it satisfies the following conditions:

- (1) $e \in H$.
- (2) H is closed under products: if $h_1, h_2 \in H$, then $h_1 h_2 \in H$.
- (3) H is closed under inverses: if $h \in H$, then $h^{-1} \in H$, where the inverse is taken in G .

A subgroup is itself a group under the same operation, with the same identity as G . Associativity is inherited from G . Recall that we also proved the following result earlier in Lecture 2.

Proposition 1.1 (Finite subgroup criterion). *If $H \subseteq G$ is finite, contains e , and is closed under products, then H is a subgroup. In this case, closure under inverses follows from the other two conditions.*

Remark. For an infinite subset, closure under inverses must still be checked. For example, $\{0, 1, 2, \dots\} \subseteq (\mathbb{Z}, +, 0)$ contains the identity and is closed under addition, but it is not closed under additive inverses.

2. GENERATED SUBGROUPS

Recall that we have also already introduced the following notion.

Definition 2.1 (Cyclic subgroup). Let G be a group, with identity denoted by 1, and let $x \in G$. The *cyclic subgroup generated by x* is

$$\langle x \rangle = \{x^m : m \in \mathbb{Z}\} = \{1, x^{\pm 1}, x^{\pm 2}, x^{\pm 3}, \dots\}. \quad (2.1)$$

We generalize this definition as follows.

Definition 2.2 (Subgroup generated by a subset). For a subset $W \subseteq G$, let $\langle W \rangle$ be the set of all finite products

$$w_1 w_2 \cdots w_k, \quad k \geq 0, \quad (2.2)$$

where each factor w_i is an element of W or the inverse of an element of W . Repetitions are allowed, and the order of the factors matters. When $k = 0$, the product is the identity 1. This is called the *subgroup generated by W* .

Indeed, concatenating two such products gives another such product. As for inverses, let's mention the following important but simple identity that you may remember from a course in linear algebra.

$$(w_1 w_2 \cdots w_k)^{-1} = w_k^{-1} \cdots w_2^{-1} w_1^{-1} \quad (2.3)$$

The product on the right is again of the required form. Hence $\langle W \rangle$ is a subgroup.

Remark. The subgroup $\langle W \rangle$ is the *smallest subgroup containing W* . Indeed, it contains W by taking products of length one. If H is any subgroup containing W , then closure under inverses and products implies that H contains every product used to define $\langle W \rangle$. Thus $\langle W \rangle \subseteq H$.

Example 2.1 (Subgroups of the integers). In the additive group $(\mathbb{Z}, +, 0)$, the even integers, $\{0\}$, and $\mathbb{Z}$ are all subgroups. More generally, for $a \in \mathbb{Z}$,

$$\mathbb{Z}a = \langle a \rangle = \{ma : m \in \mathbb{Z}\} = \{0, \pm a, \pm 2a, \pm 3a, \dots\} \quad (2.4)$$

is the subgroup consisting of all multiples of a . Here the operation is addition, so powers in multiplicative notation become integer multiples in additive notation.

Definition 2.3. For any group G , the set $\{e\}$ is a subgroup, called the *trivial subgroup*. The whole group G is also a subgroup of itself. A subgroup H is called a *proper subgroup* if $H \neq G$. If $\{e\} \subsetneq H \subsetneq G$, then H is called a *nontrivial proper subgroup*.

Intersecting subgroups provides another useful way to manufacture new groups.

Proposition 2.1. *If H and K are subgroups of G , then $H \cap K$ is a subgroup of G .*

Proof. Since $e \in H$ and $e \in K$, we have $e \in H \cap K$. If $a, b \in H \cap K$, then $ab \in H$ and $ab \in K$, because both subgroups are closed under products. Thus $ab \in H \cap K$. Similarly, if $a \in H \cap K$, then $a^{-1} \in H$ and $a^{-1} \in K$, so $a^{-1} \in H \cap K$. $\square$

Remark. The union $H \cup K$ need not be a subgroup. For example, $2\mathbb{Z} \cup 3\mathbb{Z}$ contains 2 and 3, but does not contain their sum 5.

3. SUBGROUPS OF $\mathbb{Z}$ AND GREATEST COMMON DIVISORS

Lemma 3.1. *Every subgroup of $(\mathbb{Z}, +, 0)$ is either $\{0\}$ or $\mathbb{Z}d$ for some positive integer d .*

Proof. Let $H \subseteq \mathbb{Z}$ be a subgroup with $H \neq \{0\}$. Since H contains a nonzero integer and its negative, it contains a positive integer. By the well-ordering principle, H has a smallest positive element d . We claim that $H = \mathbb{Z}d$.

First, $\mathbb{Z}d \subseteq H$, since $d \in H$ and H is closed under addition and additive inverses.

For the reverse inclusion, let $m \in H$. By division with remainder, write

$$m = qd + r, \quad q \in \mathbb{Z}, \quad 0 \leq r < d. \quad (3.1)$$

Since $m, qd \in H$, we have $r = m - qd \in H$. If $r > 0$, this contradicts the choice of d as the smallest positive element of H . Thus $r = 0$, so $m = qd \in \mathbb{Z}d$. Hence $H \subseteq \mathbb{Z}d$, proving the claim. $\square$

Greatest common divisors. Let $a, b \in \mathbb{Z}$ be not both zero. Consider the set of all integer linear combinations of a and b :

$$\mathbb{Z}a + \mathbb{Z}b = \{ma + nb : m, n \in \mathbb{Z}\}. \quad (3.2)$$

This is a subgroup of $(\mathbb{Z}, +, 0)$: it contains 0, and sums and negatives of integer linear combinations are again integer linear combinations. Therefore the preceding lemma gives

$$\mathbb{Z}a + \mathbb{Z}b = \mathbb{Z}d \quad (3.3)$$

for some $d > 0$, since this subgroup contains a nonzero element. We will show that this positive generator is $\gcd(a, b)$.

Lemma 3.2 (Divisibility and inclusion). *For integers r, s ,*

$$r \mid s \iff \mathbb{Z}s \subseteq \mathbb{Z}r. \quad (3.4)$$

Proof. If $s = kr$ for some $k \in \mathbb{Z}$, then every multiple of s is a multiple of r . Conversely, if $\mathbb{Z}s \subseteq \mathbb{Z}r$, then $s \in \mathbb{Z}r$, so $r \mid s$. $\square$

Example 3.1. The direction of inclusion reverses the order in the divisibility relation:

$$2 \mid 6, \quad 6\mathbb{Z} \subseteq 2\mathbb{Z}. \quad (3.5)$$

Indeed, every multiple of 6 is a multiple of 2.

Proposition 3.1. *Let $a, b \in \mathbb{Z}$ be not both zero, and let $d > 0$ satisfy $\mathbb{Z}a + \mathbb{Z}b = \mathbb{Z}d$. Then $d = \gcd(a, b)$. In particular,*

$$\mathbb{Z}a + \mathbb{Z}b = \mathbb{Z}\gcd(a, b). \quad (3.6)$$

Proof. We check two properties:

- (1) d divides both a and b .
- (2) Every common divisor of a and b divides d .

For the first property, the inclusions

$$\mathbb{Z}a \subseteq \mathbb{Z}a + \mathbb{Z}b = \mathbb{Z}d, \quad \mathbb{Z}b \subseteq \mathbb{Z}a + \mathbb{Z}b = \mathbb{Z}d \quad (3.7)$$

imply $d \mid a$ and $d \mid b$ by Lemma 3.2.

For the second property, suppose $\ell \mid a$ and $\ell \mid b$. Then $\mathbb{Z}a \subseteq \mathbb{Z}\ell$ and $\mathbb{Z}b \subseteq \mathbb{Z}\ell$. Adding elements from these two subgroups gives

$$\mathbb{Z}d = \mathbb{Z}a + \mathbb{Z}b \subseteq \mathbb{Z}\ell. \quad (3.8)$$

By Lemma 3.2, $\ell \mid d$. Every positive common divisor ℓ divides d , so $\ell \leq d$. Since d itself divides both a and b , it is their greatest common divisor. $\square$

This gives Bézout's identity.

Proposition 3.2 (Bézout's identity). *Let $a, b \in \mathbb{Z}$ be not both zero. There exist integers m, n such that*

$$\gcd(a, b) = ma + nb. \quad (3.9)$$

Proof. Set $d = \gcd(a, b)$. Since $\mathbb{Z}a + \mathbb{Z}b = \mathbb{Z}d$ and $d \in \mathbb{Z}d$, we have $d \in \mathbb{Z}a + \mathbb{Z}b$. By the definition of this subgroup, $d = ma + nb$ for some $m, n \in \mathbb{Z}$. $\square$